



Technical Reproducibility Validation Report

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y

Version v0.7

Y-STR

Verification date	2026-07-19
Repository commit	b2033bfb5cf55496b776463bdf2993fa763a4be
Attestation level	Installs
Permalink	https://strhub.app/verified/ztest-upload-inventado-slug-ztest-upload-v0-7-y-v0-7-y
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	2/5 gates passed — Installs
ATTESTATION LEVEL	Installs
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y
Version	v0.7
Repository	https://github.com/tfwillems/HipSTR
Commit (pinned)	b2033bfb5cf55496b776463bdf2993fa763a4be
Container OS	ubuntu-22.04
Marker panel	Y-STR
Verified on	2026-07-19 18:00:37 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/29697849727

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y v0.7 · commit
b2033bf
$ HipSTR \
--bams \
/data/in/input.bam \
--fasta \
/data/ref/hg38.fa \
--regions \
/data/in/regions.bed \
--str-vcf \
/data/out/result.vcf.gz \
--min-reads \
5 \
--use-unpaired \
--read-qual-trim \
! \
--def-stutter-model \
--max-str-len \
250
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	—
Expected IO	Produces a non-empty file in the declared format	—
Output Structure Validation	Output structure matches expected genotype-bearing format	—
Execution log	ztest-upload-inventado-slug-ztest-upload-v0-7-y-v0-7-y.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	result.vcf.gz
Format	VCF
Sequence records	0
Distinct STR loci	0
Total reads	0
Max depth (single locus)	0

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 14 forensic STR loci, not a whole genome: it carries reads only at the loci listed below.

Dataset	Illumina BAM (hg38) — HG002 (Y-STR, male)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38

Loci tested	—
Regions BED	Provided by the tool author — covers 14 of 14 supported loci

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38) — HG002 (Y-STR, male)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type – Single containerized environment (Docker / ubuntu-22.04) – No truth-set comparison or ground-truth genotypes – No accuracy or concordance assessment – No forensic validation of results – Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation. Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ Runs end-to-end

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 2 verification gates were passed.

◆ Produces valid output

The tool generated a structurally valid VCF output file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

◆ No bundled demo data

Ztest-Upload (Inventado → Slug Ztest-Upload-V0-7-Y does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38) — HG002 (Y-STR, male) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.